Exam 2

Graded

Student

Myles Bonham

Total Points

87.25 / 100 pts

Question 1

Question 1

3 / 3 pts

✔ **– 0 pts** Correct

Question 2

Question 2

3 / 3 pts

✔ **– 0 pts** Correct

Question 3

Question 3

3 / 3 pts

✔ **– 0 pts** correct

Question 4

Question 4

3 / 3 pts

✔ **– 0 pts** Correct

Question 5

Question 5

3 / 3 pts

✔ **– 0 pts** Correct

Question 6

Question 6

0 / 3 pts

✔ **– 3 pts** Incorrect

Question 7

Question 7

1.5 / 3 pts

✔ **– 1.5 pts** One correct selection

Question 8

Question 8 **11** / 13 pts

8.1 Question 8a **7** / 7 pts

✔ **− 0 pts** Correct

8.2 Question 8b **4** / 4 pts

✔ **− 0 pts** Correct

8.3 Question 8c **0** / 2 pts

✔ **− 2 pts** Incorrect. The correct answer is $\mathbb{R}^4$.

Question 9

Question 9 **20** / 20 pts

9.1 Question 9a **4** / 4 pts

✔ **− 0 pts** Correct

9.2 Question 9b **4** / 4 pts

✔ **− 0 pts** Correct

9.3 Question 9c **12** / 12 pts

✔ **− 0 pts** Correct

Question 10

Question 10

10 / 10 pts

✔ **+ 10 pts** Correct

+ 0 pts blank

+ 5 pts Correctly identified the matrix

$$\begin{vmatrix} 4 & 4 & 2 \\ 2 & 0 & 2 \\ 2 & 3 & 0 \end{vmatrix}$$

For using cramer’s rule

+ 2 pts Student did not replace the x column with vector b.
It should be:

$$\begin{vmatrix} 4 & 4 & 2 \\ 2 & 0 & 2 \\ 2 & 3 & 0 \end{vmatrix}$$

+ 3 pts correctly calculated the determinant

+ 2.5 pts determinant is very close to correct (minor error)

+ 2 pts error(s) in finding determinant

+ 1 pt incorrect/unclear procedure for calculating the determinant

+ 0 pts determinant not found

+ 1 pt divided by 32

+ 0 pts did not divide by 32 (given in problem)

+ 1 pt answer provided

+ 5 pts Student solved for x without using Cramer's rule

+ 3 pts Student attempted to solve for x without using Cramer's rule

+ 2 pts minimal work shown

Question 11

Question 11

3 / 8 pts

✔ **− 5 pts** Several Matrix multiplication errors

1 $$\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} C & D \\ A & B \end{bmatrix}$$

Question 12

Question 12

11 / 12 pts

+ 12 pts Fully correct

✔ **+ 2 pts** attempt made to find answer

✔ **+ 1 pt** answer follows from work

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matrix L (max 5 points)

✔ **+ 2 pts** lower triangular

+ 1 pt almost lower triangular

+ 0 pts not lower triangular

✔ **+ 1 pt** 1's on diagonal (student might have applied to upper matrix instead)

+ 0 pts In LU factorization, the diagonal of (usually) L should be 1's in order to guarantee a unique solution.

+ 2 pts correct values

+ 1.5 pts almost correct values

✔ **+ 1 pt** some incorrect values

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matrix U (max 4 points)

✔ **+ 2 pts** upper triangular

+ 1 pt almost upper triangular

+ 0 pts not upper triangular

✔ **+ 2 pts** correct values

+ 1.5 pts almost correct values

+ 1 pt some incorrect values

+ 0 pts no meaningful work

swapped L and U in answer (no credit lost)

Question 13

Question 13

6 / 6 pts

✔ **− 0 pts** Correct

Question 14

Question 14 **9.75** / 10 pts

14.1 Question 14a **6** / 6 pts

- **− 0 pts** Correct

14.2 Question 14b **3.75** / 4 pts

- **− 0.25 pts** Minor arithmetic error/algebraic error

2 2

Name: Myles Bonham   Student ID: 12626000

Math 3108 – Linear Algebra
Exam 2
Prof. Runnion
April 4, 2024

Directions:

1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
2) No calculators, books, or other materials are permitted.
3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
5) Failure to follow directions specific to a problem will result in the loss of points.
6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
7) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
8) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
9) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos\pi$ should be calculated.
10) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!
11) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
12) This packet has seven sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
13) Once this exam begins, you will have 50 minutes to complete your solutions.

DO NOT OPEN THIS EXAM UNTIL TOLD TO DO SO

Name:__________________________________ **Student ID:**____________________

Multiple Choice and True/False. Fill in the box corresponding to the correct answer.
3 points each.

1. A basis for a vector space V is a __________ of vectors in V.

   ☐ i) minimal spanning set ☐ ii) maximal linearly independent set

   ☒ iii) both (a) and (b) ☐ iv) neither (a) nor (b)

2. For an $n \times n$ matrix A, if the equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then the columns of A span $\mathbb{R}^n$.

   ☐ i) Always true ☐ ii) Sometimes true ☒ iii) Never true

3. If there is an $n \times n$ matrix D such that $AD = I$, then the columns of A^{-1} are linearly independent.

   $A\ A^{-1}$

   ☒ i) Always true ☐ ii) Sometimes true ☐ iii) Never true

4. If A and B are invertible $n \times n$ matrices, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$

   ☒ i) Always true ☐ ii) Sometimes true ☐ iii) Never true

5. The number of free variables in the equation $A\mathbf{x} = \mathbf{0}$ equals the rank of A.

   ☐ i) Always true ☒ ii) Sometimes true ☐ iii) Never true

6. If A is a 3×7 matrix, what is the smallest possible dimension of Nul A?

   $A\mathbf{x} = 0$

   ☒ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9

7. Let A be an $m \times n$ matrix. Which of the following is a subspace of $\mathbb{R}^n$?
   Select all that apply.

   ☒ i) Col A (the column space of A)

   ☒ ii) Nul A (the null space of A)

   ☐ iii) Row A (the row space of A)

Name:__________________________________ **Student ID:**____________________

Open Answer. Show ALL your work! Place your final answer in the box provided, if applicable.

8. Consider the vector space $H = \left\{ \begin{bmatrix} 12a+24b-4c \\ 6a-2b-2c \\ -9a+5b+3c \\ -3a+b+c \end{bmatrix} : a,b,c \text{ real} \right\}$.

a) (7 points) Find a basis for H.

$\begin{bmatrix} 12 \\ 6 \\ -9 \\ -3 \end{bmatrix} a + \begin{bmatrix} 24 \\ -2 \\ 5 \\ 1 \end{bmatrix} b + \begin{bmatrix} -4 \\ -2 \\ 3 \\ 1 \end{bmatrix} c$ Not lin indep

$-\frac{1}{3}\begin{bmatrix} 12 \\ 6 \\ -9 \\ -3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ -3 \\ -1 \end{bmatrix}$ & $\begin{bmatrix} 12 \\ 6 \\ -9 \\ -3 \end{bmatrix} \neq \begin{bmatrix} 24 \\ -2 \\ 5 \\ 1 \end{bmatrix}$

Basis $\begin{bmatrix} 12 \\ 6 \\ -9 \\ -3 \end{bmatrix}, \begin{bmatrix} 24 \\ -2 \\ 5 \\ 1 \end{bmatrix}$

b) (4 points) The dimension of H is $\boxed{2}$.

c) (2 points) The set ~~W~~ H is a subspace of. (Fill in the box.)

☒ i) $\mathbb{R}^2$ ☐ ii) $\mathbb{R}^3$ ☐ iii) $\mathbb{R}^4$ ☐ iv) None of these.

Name:______________________________ **Student ID:**____________________

9. The matrices A and B given below are row equivalent. (You do *not* need to check this.)

$$A = \begin{bmatrix} -2 & 8 & -2 & 8 \\ 2 & -12 & -4 & 1 \\ -3 & 16 & 3 & -5 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & 7 & 11 \\ 0 & 4 & 6 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

a) (4 points) Find a basis for the column space of A.

$$\begin{bmatrix} -2 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 8 \\ -12 \\ 16 \end{bmatrix}$$

b) (4 points) Find a basis for the row space of A.

$$[1\ 0\ 7\ 11], \quad [0\ 4\ 6\ 7]$$

c) (12 points) Find a basis for the null space of A.

$AX = 0$ $\quad x_1 = -7x_3 - 11x_4 \quad x_2 = -\frac{3x_3}{2} - \frac{7x_4}{4}$

$$\left[\begin{array}{cccc|c} 1 & 0 & 7 & 11 & 0 \\ 0 & 4 & 6 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{bmatrix} -7 \\ -3/2 \\ 1 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} -11 \\ -7/4 \\ 0 \\ 1 \end{bmatrix} x_4$$

$$\begin{bmatrix} -7 \\ -3/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -11 \\ -7/4 \\ 0 \\ 1 \end{bmatrix}$$

Basis for Nul A

Name:____________________ **Student ID:**________________

10. (10 points) Consider the system of equations $\begin{cases} 6x+4y+2z=4 \\ 6x \qquad\; +2z=2 \\ 4x+3y \qquad\; =2 \end{cases}$.

Given that $\begin{vmatrix} 6 & 4 & 2 \\ 6 & 0 & 2 \\ 4 & 3 & 0 \end{vmatrix} = 32$, use Cramer's Rule to solve for the variable x only.

$D=$

$$D_x = \begin{vmatrix} 4 & 4 & 2 \\ 2 & 0 & 2 \\ 2 & 3 & 0 \end{vmatrix} \underset{-\frac{1}{2}R_1+R_3\to R_3}{\overset{-\frac{1}{2}R_1+R_2\to R_2}{=}} \begin{vmatrix} 4 & 4 & 2 \\ 0 & -2 & 1 \\ 0 & 1 & -1 \end{vmatrix} = 4\begin{vmatrix} -2 & 1 \\ 1 & -1 \end{vmatrix} -0+0$$

$4((-2)(-1)-(1)(1))$

$4(2-1)$

4

$D_x = 4$

$$X = \frac{D_x}{D} = \frac{4}{32} = \frac{1}{8}$$

$\frac{\cancel{4}\,1}{\cancel{4}\cdot 8}$

Solution: $x = \boxed{1/8}$

11. (8 points) Assume that the matrices shown below are partitioned conformably for block multiplication. Compute the product.

$$\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 0-B & A+0 \\ 0-D & C+0 \end{bmatrix}$$

Solution: $\begin{bmatrix} -B & A \\ -D & C \end{bmatrix}$

Name:________________________________ **Student ID:**____________________

12. (12 points) Find the LU factorization of the matrix $A = \begin{bmatrix} 2 & -4 & 2 \\ 2 & -2 & 1 \\ -1 & 4 & 0 \end{bmatrix}$.

$$\xrightarrow[\frac{1}{2}R_1 + R_3 \to R_3]{-R_1 + R_2 \to R_2}$$

$$\begin{bmatrix} 2 & -4 & 2 \\ 0 & 2 & -1 \\ 0 & 2 & 1 \end{bmatrix} \xrightarrow{-R_2 + R_3 \to R_3}$$

$$U = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -\frac{1}{2} & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -4 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

Solution: $L = \begin{bmatrix} 2 & -4 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{bmatrix}$ $U = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ \frac{1}{2} & & 1 \end{bmatrix}$

13. (6 points) Show that the set of all polynomials of the form $\mathbf{p}(t) = a + bt$ where a and b are real is closed with respect to scalar multiplication.

$$P_1(t) = a_1 + b_1 t$$

$$c\,P_1(t) = \underbrace{ca_1}_{\text{(a is scalar)}} + \underbrace{c\,b_1}_{\text{(b is scalar)}} t$$

Still in Form $p(t) = a + bt$

both Real as long as c is real

$\therefore$ $P(t)$ is closed with respect to Scalar multiplication

Name:____________________________________ **Student ID:**____________________

14. a) (6 points) Use the method of your choice to calculate A^{-1} given $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$.

$$\frac{1}{(2)(2)-(1)(3)} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

$4 - 3 = 1$

Solution: $A^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$

b) (4 points) Use the inverse you found in part (a) to solve $A\mathbf{x} = \mathbf{b}$ given $\mathbf{b} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$.

$\cancel{A^{-1}AX = bA^{-1}}$

$X = A^{-1}b$

$$\underset{2\times 2}{\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}} \underset{2\times 1}{\begin{bmatrix} 3 \\ 2 \end{bmatrix}} = \begin{bmatrix} \overset{6 \;\; +3 \;=\; 9}{2(3)-(-1)(2)} \\ \underset{-9 \;-\; 4 \;=\; -13}{(-3)(3)-(2)(2)} \end{bmatrix}$$

Solution: $\mathbf{x} = \begin{bmatrix} 9 \\ -13 \end{bmatrix}$